Indian Statistical Institute
BSDS Semester-I 2026-2027
Subject: Introduction to Computation
Class Test 1 (12 November, 2026)
Maximum marks: 20 Duration: 1 hr.

1. (a) Let A be a matrix that is stored as a list of lists (all of which are of the same length). Write code to create A^T the transpose of A (A^T should also be stored as a list of lists). For full credit, your code should be a one-line, list comprehension expression. [1]

   (b) Suppose that the entries of A are all strings. Write code to create a string that is obtained by concatenating all the entries of A in row-major order. For full credit, your code should be a one-line, list comprehension expression. [2]

   (c) Show how you would modify your solution to (b) if you were required to concatenate **only single character strings**, in **column-major** order. For full credit, your code should be a one-line, list comprehension expression. [3]

   After we have covered regular expressions in class, you should take a look at `https://www.hackerrank.com/challenges/matrix-script/problem`.

2. Write a Python function `merge_dict(d1, d2)` that takes two dictionaries `d1`, `d2` as arguments, and returns a dictionary that contains all keys in `d1` and `d2`. If `d1` and `d2`, share any keys, the value for such keys in the result should be the sum of the corresponding values from `d1` and `d2` (you may assume these are both integers). For example, if `dict1 = {"a": 10, "b": 20, "c": 30}` and `dict2 = {"b": 5, "c": 15, "d": 25}`, your function should return `{"a": 10, "b": 25, "c": 45, "d": 25}`. The order of the keys in the result is not important. [4]

P.T.O.

3. Recall that if $f(x)$ is continuous over $[a, b]$

$$\int_a^b f(x)dx \;=\; \lim_{n\to\infty} \sum_{i=0}^{n-1} f\big(a + \frac{i}{n}(b-a)\big) \times \frac{b-a}{n}.$$

Write a Python function `integrate(f, a, b)` that uses the above idea to compute $\int_a^b f(x)dx$. Use a fixed value of $n = 10^3$ in your code. [4]

You may use NumPy in your solution if you like, but plain Python solutions are also acceptable.

4. A number is said to be a *Disarium Number* if the sum of its digits raised to their respective positions is the number itself. The position of the leftmost significant digit is defined as 1; subsequent digits to the right are assigned positions 2, 3, ….. For example, 135 and 518 are Disarium Numbers, since $1^1+3^2+5^3 = 1+9+125 = 135$, and $5^1 + 1^2 + 8^3 = 518$. Single digit numbers are all Disarium Numbers, but 75 ($7^1 + 5^2 = 7 + 25 = 32$) and 544 are not.

Write a Python function `is_disarium(n)` that returns `True` if `n` is a Disarium Number, and `False` otherwise. [6]